The Clebsch cubic — **III**

Golden descent and operator realizations of the Clebsch cubic

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From deep holes, the cubic takes shape, finds its bearings,
stands fixed while its shadows move*, rebuilds the plane that cast it,*
and gathers its shadows home.

Abstract

Let H be the rational seven-space of harmonic cubics. Hitchin's icosahedral incidence variety is generically a degree-two cover of $\mathbf{P}(H)$. We prove that its function field is $\mathbf{Q}(\mathbf{P}(H))(\sqrt{5J_0})$, where J_0 is the rational equation of the reduced branch sextic normalized by $\iota_t^* J_0 = 16\sigma_3^2$ on the Clebsch chart. The complete reduced fibre over $[xyz]$ has residue algebra $\mathbf{Q}(\sqrt{5})$ and determines the twist; the finite Stein algebra is $\mathcal{O} \oplus \mathcal{O}(-3)$ with multiplication $z^2 = 5J_0$.

After an ordering, chart lift, outer labels, and Petersen labels are fixed as a marked bridge datum, a chosen sheet selects an order-six conference source or its opposite. The sheet alone supplies none of that marking. The source cubic and its six outer translates have four exact descriptions: triangle holonomy, middle-exterior diagonal, commutator Pfaffian, and oriented cross-golden determinant. They give the signed Joubert–Segre–Igusa–Clebsch chain. On the Petersen four-space the degree-six zonal-harmonic cubic restricts exactly to $-784000\sigma_3/1247103$. This is a relative sign comparison between cubics on different spaces, not an identification of their ambient harmonic representations.

The same conference carrier has two independent structural consequences. Cut-independence of the balanced exchange spectrum singles out order six. Aligned four-sets reconstruct every two-graph on at least seven vertices up to complement, with seven sharp; hence the determinant-(-3) blocks recover symmetric conference signings of order at least ten up to switching and global negation.

Keywords. Clebsch configuration; Hitchin cover; icosahedral invariant; Gaunt coefficients; symmetric conference matrix; two-graph; Seidel matrix; golden ratio.

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1 Introduction

Let

$$H = \ker\Big(\partial_x^2 + \partial_y^2 + \partial_z^2 : \mathbf{Q}[x,y,z]_3 \longrightarrow \mathbf{Q}[x,y,z]_1\Big)$$

be the rational seven-space of harmonic cubics for $Q(x,y,z) = x^2 + y^2 + z^2$. We use the polarization pairing on cubics normalized by

$$\langle p, (\,a \cdot x\,)^3 \rangle = p(a);$$

its restriction to H is nondegenerate. Hitchin associated to a general $[f] \in \mathbf{P}(H)$ two icosahedral six-axis configurations: equivalently, two Mukai–Umemura isotropic three-planes $U \subset H$ with $f \perp U$ [1, 2]. The incidence variety of pairs $([f], U)$ is therefore generically a degree-two cover of $\mathbf{P}(H)$. We determine its rational square class and, after fixing the marked bridge datum below, compare its chosen sign with two further realizations. On the six-axis carrier the marked sign determines a conference operator whose Pfaffian, determinant, and middle-exterior shadows give the classical Joubert, Segre, Igusa, and diagonal Clebsch models. On the ten face axes it returns as the degree-six Gaunt cubic. Figure 1 records this architecture and the role of the marking.

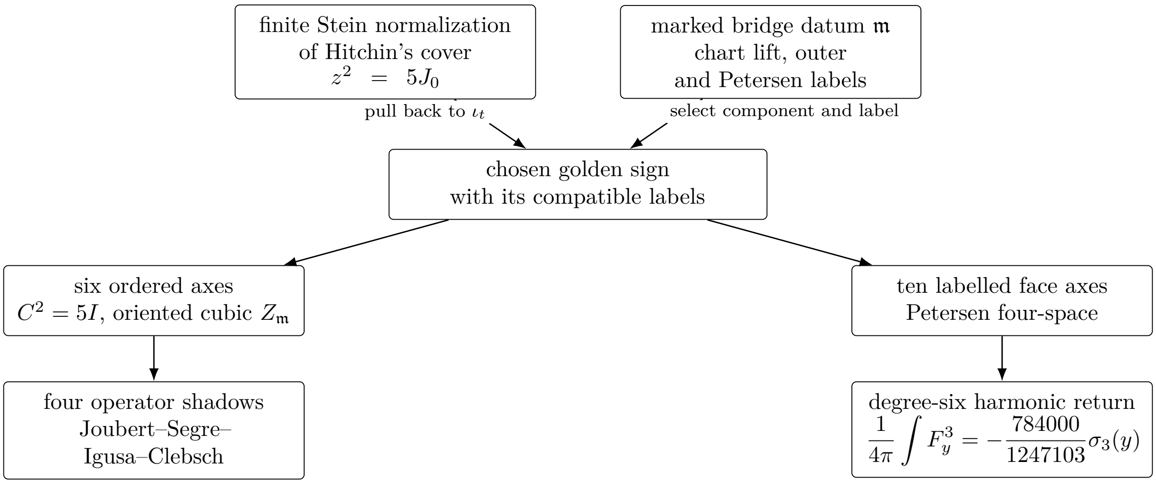

Figure 1: The source–shadow–return scheme. Solid arrows are constructions after the two top inputs have been fixed. The lower branches compare the chosen sign; they neither identify the displayed cubics nor define a map between their ambient harmonic spaces.

First-pass route. The main argument runs through the incidence cover in Section 2, its marked pullback in Section 3, the golden fibre and exchanger in Sections 4.1 and 4.2, the four operator shadows in Section 5.1, and the harmonic return in Section 6. The finite-field specialization in Section 4.3 and the exchange-spectrum and reconstruction subsections of Section 5 are independent consequences and may be skipped on a first reading.

Hitchin defines a real invariant sextic from the characteristic polynomial of a quadratic moment operator. His appendix rescales that operator while computing its restriction, so we separate the geometric hypersurface from its scale. Let J_0 denote the rational equation of Hitchin's irreducible sextic hypersurface with the exact normalization

$$\iota_t^* J_0 = 16\sigma_3^2$$

for the lifted Clebsch chart defined in Section 2. The rationality and uniqueness of this normalization, and its relation to Hitchin's real invariant, are checked there [1, opening theorem and Appendix]. Write

$$V = \{(y_1, \dots, y_5) : \textstyle\sum_i y_i = 0\}, \qquad \sigma_3(y) = \frac{1}{3}\textstyle\sum_i y_i^3$$

for the Clebsch four-space and its invariant cubic. Here and below, $D(\sigma_3)$ is the principal open set $\{\sigma_3 \neq 0\}$. Because J_0 is a section of $\mathcal{O}_{\mathbf{P}(H)}(6)$, we use

$$K(\sqrt{cJ_0}), \qquad K = \mathbf{Q}(\mathbf{P}(H)),$$

as shorthand: choose a nonzero rational section s of $\mathcal{O}_{\mathbf{P}(H)}(3)$, put $j_s = J_0/s^2 \in K^\times$, and take $K(\sqrt{cj_s})$. Replacing s changes j_s by a square, so the quadratic extension is unchanged.

Theorem 1.1 (Arithmetic incidence cover)**.** *The function field of Hitchin's incidence cover is*

$$\mathbf{Q}(\mathbf{P}(H))(\sqrt{5J_0}).$$

If $\mathcal{N} \to \mathbf{P}(H)$ is its finite Stein normalization, then, after normalizing the anti-invariant summand, its algebra is

$$\mathcal{O} \oplus \mathcal{O}(-3), \qquad z^2 = 5J_0.$$

For the Clebsch chart $\iota_t : \mathbf{P}(V_{\mathbf{Q}(\sqrt{5})}) \to \mathbf{P}(H_{\mathbf{Q}(\sqrt{5})})$ defined in Section 2,

$$\iota_t^* J_0 = 16\sigma_3^2.$$

The chart is not defined over $\mathbf{Q}$; Galois conjugation carries it to the chart of the conjugate icosahedron. The fibre over the rational point $[xyz]$ is nevertheless a complete reduced fibre with residue algebra $\mathbf{Q}(\sqrt{5})$.

The underlying Galois exchange of the two conjugate icosahedra is classical [13, Section 1.3]. The theorem identifies its occurrence here with the square class of Hitchin's incidence cover.

The factor 5 has a concrete descent meaning: the cubic $[xyz]$ and its unordered geometric fibre are rational, but choosing either icosahedral configuration—equivalently, either conjugate Clebsch chart through that point—requires $\mathbf{Q}(\sqrt{5})$. This fibre is the model case for the descent: its residue algebra determines the global quadratic twist in the proof of Theorem 1.1.

The choice can be retained globally after normalizing the pullback to the Clebsch chart, but its comparison with the conference and harmonic signs is relative to a marking. A *marked bridge datum* $\mathfrak{m}$ consists of an ordering of the six golden projective axes, the five plane triples labelled so that $q_1 = xyz$ and $\sum q_i = 0$, and the resulting normalized linear lift

$$\iota_t(y) = \sum_i y_i q_i.$$

The datum also identifies the ten face axes with the two-subsets of those five labels. Choose normalized representatives of the six projective axes. They give a conference matrix $C_\mathfrak{m}$. A different choice switches it by a diagonal sign matrix. The resulting switching-invariant triangle cubic is

$$Z_\mathfrak{m}(x) = \sum_{i<j<k} (C_\mathfrak{m})_{ij}(C_\mathfrak{m})_{jk}(C_\mathfrak{m})_{ki} x_i x_j x_k.$$

It is translation-invariant and hence defines

$$Z_\mathfrak{m} \in \operatorname{Sym}^3((\mathbf{Q}^6/\mathbf{Q}\mathbf{1})^*).$$

Write $[C_\mathfrak{m}, Z_\mathfrak{m}]_{\mathrm{or}}$ for this marked source. Changing the auxiliary representatives by $a_i \mapsto d_i a_i$ sends $C_\mathfrak{m}$ to $DC_\mathfrak{m}D$ and fixes $Z_\mathfrak{m}$. The opposite source is represented by $(-C_\mathfrak{m}, -Z_\mathfrak{m})$. The sign of the cubic generator is retained, so the brackets are not projective notation. Relabelling is equivariant only when all labelled objects are transported together; it is not quotiented silently.

Proposition 1.2 (Relative marked orientation bridge)**.** *Fix a marked bridge datum $\mathfrak{m}$. Let $B = \mathbf{P}(V_E)$, where $E = \mathbf{Q}(\sqrt{5})$, and pull Hitchin's incidence normalization back along $\iota_t : B \to \mathbf{P}(H_E)$. The normalization of this pullback is a disjoint union*

$$B_+ \amalg B_-.$$

The fixed isotropic plane selects B_+, and incidence deck exchange interchanges the components. For an odd generator normalized on B_+, their equations are

$$z = 4\sqrt{5}\,\sigma_3, \qquad z = -4\sqrt{5}\,\sigma_3.$$

On $D(\sigma_3)$ they are the two distinct icosahedral configurations; on the Clebsch cubic the unnormalized branches meet, while the normalization remains disjoint.

Relative to $\mathfrak{m}$, equip B_+ with the source $[C_\mathfrak{m}, Z_\mathfrak{m}]_{\mathrm{or}}$ and equip B_- with its deck-opposite. At $[xyz]$, the golden exchanger identifies this opposite with the conjugate configuration and the representative $(-C_\mathfrak{m}, -Z_\mathfrak{m})$. Under the marked primitive pair-sum map

$$\beta : V \longrightarrow \mathbf{Q}^{\binom{5}{2}}, \qquad \beta(y)_{ij} = y_i + y_j,$$

the same relative convention fixes the sign of the degree-six cubic in Theorem 6.1. The normalized component alone is not asserted to determine $\mathfrak{m}$.

Here $Z_\mathfrak{m}$ and σ_3 are different cubics on spaces of dimensions five and four. The proposition compares their orientations relative to $\mathfrak{m}$; it does not identify their polynomial domains.

Theorem 5.1 is the operator stage on the main route. The six outer translates of $Z_\mathfrak{m}$ are simultaneously triangle cubics, middle-exterior diagonals, commutator Pfaffians, and cross-golden determinants. Their Joubert coordinates land on the Segre cubic, whose coordinate sections are diagonal Clebsch cubic surfaces; centered squaring is the classical Segre–Igusa polar map [13, Sections 2.1–2.2]. The marked datum defines this operator family.

The proof follows the language changes in Figure 1. First, the rational incidence model and its canonical classes give a quadratic field branched exactly along the reduced sextic; the complete etale fibre at $[xyz]$ then determines its square class [5]. Next, Proposition 1.2 normalizes the chart pullback and, relative to the stated marking and lift, follows deck exchange through the golden conference source and Petersen pair-sum map. Finally, Theorem 5.1 identifies the four operator formulas, and Theorem 6.1 identifies the Petersen coefficient module and computes the invariant cubic by one exact spherical-moment evaluation.

Two later subsections of Section 5 use the same conference carrier but supply no hypothesis to this route. Theorem 5.2 characterizes order six by cut-independent balanced exchange spectrum. Independently, Theorem 5.3 proves that aligned four-sets reconstruct two-graphs from seven vertices onward, with seven sharp. For conference matrices the latter recovers the signing from order ten onward, up to switching and global negation.

Hitchin proves the generic degree, identifies the branch sextic, constructs the Clebsch chart, computes its restriction, and classifies the fibre over xyz [1, §§2–4, 7–10] [2, §§4–5]. The arithmetic theorem fixes the remaining rational constant in that square class and computes the spinor class of the explicit specialization. The Mukai–Umemura threefold supplies the original threefold [4]; Hitchin supplies the incidence construction and every formula used here. Dye's finite theorem gives the complementary square-5 criterion for Clebsch hexagons [3]. The abstract equation $z^2 = 5J_\mathbf{Z}$ is etale on $D(10J_\mathbf{Z})$, but its comparison with the geometric incidence variety is proved only after inverting an unspecified integer.

The harmonic theorem places the classical degree-six bond-order observable—the rotationally invariant cubic formed from degree-six harmonic coefficients— [6, 7] on the same four-space and computes its exact restriction. The coefficient factors as

$$-\frac{784000}{1247103} = -\frac{400}{46189}\frac{1960}{27}.$$

The first factor is the universal $(6,6,6)$ Gaunt coupling $\begin{pmatrix} 6 & 6 & 6 \\ 0 & 0 & 0 \end{pmatrix}^2$; the second is the restriction scalar of the marked Petersen embedding. Thus the prime 11 in the denominator belongs to the universal angular momentum normalization, not to the golden descent. The relative comparison uses their four-dimensional A_5-modules and the invariant polynomial σ_3.

The decomposition of spherical harmonics under icosahedral symmetry is classical, including the degree-six symmetry types used here [9, 10]; modern applications likewise treat degree six as the lowest nonconstant even icosahedral channel [11]. Our calculation fixes a particular labelled ten-face-axis zonal map, identifies its Petersen four-space, and evaluates the cubic on that map. It does not claim priority for the ambient icosahedral decomposition or for the bond-order invariant.

2 The rational incidence cover

2.1 The Clebsch chart

Put $E = \mathbf{Q}[t]/(t^2 - t - 1) = \mathbf{Q}(\sqrt{5})$, and let

$$I_t = \{[0 : t : 1], [0 : t : -1], [1 : 0 : t], [-1 : 0 : t],\\ [t : -1 : 0], [-t : -1 : 0]\} \subset \mathbf{P}^2_E.$$

These are the six axes of the standard icosahedron. Define the linear subspace

$$V_t = \{p \in H_E : p(a) = 0 \text{ for every } [a] \in I_t\} \subset H_E.$$

This is the exact inclusion used below. Hitchin proves that V_t has dimension four and describes it as follows [1, §§2–4]. The rotation group of I_t permutes the five triples of mutually orthogonal symmetry planes. Let $q_1, \ldots, q_5$ be the products of the three linear equations in those triples, with $q_1 = xyz$ and relative scalars chosen so that $\sum_i q_i = 0$. Then

$$\iota_t : V_E = \{(y_i) \in E^5 : \textstyle\sum_i y_i = 0\} \longrightarrow V_t, \qquad (y_i) \longmapsto \textstyle\sum_i y_i q_i$$

is an A_5-equivariant isomorphism. Thus the parameter of $xyz = q_1$ is

$$y^\circ = \frac{1}{5}(4, -1, -1, -1, -1), \qquad \sigma_3(y^\circ) = \frac{4}{25}.$$

We first make the sextic scale precise. For a real harmonic cubic f, divide Hitchin's symmetric moment operator by 4π and write

$$M_f = \frac{1}{4\pi} \int_{S^2} f(x)^2 \left(xx^\mathsf{T} - \frac{4}{15} I \right) d\omega$$

Hitchin applies $3a_3 - 4a_1a_2$ to the coefficients of its characteristic polynomial $\lambda^3 + a_1\lambda^2 + a_2\lambda + a_3$. Normalized spherical moments of monomials are rational, so this is a rational sextic polynomial on H. Its zero set is Hitchin's sextic. We record its irreducibility paper-locally. After complexification, let $S_\mathbf{C} \subset \mathbf{P}(V_\mathbf{C})$ be the Clebsch diagonal cubic surface. It is smooth, hence irreducible, and $\mathrm{SO}(3, \mathbf{C})$ is connected, hence irreducible. Therefore the Zariski closure of the image of the action morphism

$$\mathrm{SO}(3, \mathbf{C}) \times S_\mathbf{C} \longrightarrow \mathbf{P}(H_\mathbf{C})$$

is irreducible. Hitchin's orbit description identifies this closure with the sextic hypersurface [1, §9]. Thus the hypersurface, and hence its rational scalar normalization $J_0 = 0$ below, is irreducible over $\mathbf{C}$. In the appendix Hitchin rescales the moment operator before the final restriction calculation; we therefore use the opening formula only to fix the hypersurface, not to identify the two printed scales.

The pullback of any rational equation of this hypersurface is a nonzero scalar multiple of σ_3^2. That scalar lies in $\mathbf{Q}$: applying $t \mapsto 1-t$ replaces the chart by its conjugate without changing either the rational equation or σ_3^2. There is consequently a unique rational multiple, denoted J_0, for which Hitchin's appendix calculation [1, Appendix, final displayed formula] reads

$$\iota_t^* J_0 = 16\sigma_3^2, \qquad J_0(xyz) = \left(\frac{16}{25}\right)^2.$$

This is an internal rational normalization, not an identification of two analytic scales in the source. The coefficient is fixed by the chosen lift: at y°, the ratio of $J_0(xyz)$ to $\sigma_3(y^\circ)^2$ is exactly 16. Over $\mathbf{R}$, J_0 is a positive multiple of Hitchin's opening invariant, so his sign classification is unchanged.

This chart is defined over E, not over $\mathbf{Q}$. The nontrivial automorphism $t \mapsto 1-t$ carries I_t and V_t to the distinct conjugate icosahedron and its vanishing space; the matrix R in Section 4 gives the corresponding transport. Hence ι_t is not a literal rational inclusion $V \hookrightarrow H$. The two displayed icosahedra have no common axis. Hitchin's two-icosahedron intersection theorem therefore gives

$$V_t \cap V_{1-t} = E \cdot xyz[1, \mathit{Proposition}\ 7].$$

Consequently the reduced union $\mathbf{P}(V_t) \cup \mathbf{P}(V_{1-t})$ descends to $\mathbf{Q}$: over E it is a pair of conjugate projective three-spaces meeting only at the rational point $[xyz]$. Denote the descended scheme by Z. Its normalization map is $\nu : Z^\nu \to Z$, where $Z^\nu = \mathbf{P}(V_t)$, regarded as a $\mathbf{Q}$-scheme through $\operatorname{Spec} E \to \operatorname{Spec} \mathbf{Q}$; after base change to E, this normalization is the disjoint union of the two conjugate components. The inverse image of the unique intersection point is

$$\nu^{-1}([xyz]) = \operatorname{Spec} E.$$

The local algebra is an instance of a general pinching construction.

Quadratic pinching.

Let E/k be a separable quadratic field extension, put $A = E[[x_1, \ldots, x_n]]$ with $n \geq 1$, and let $\mathfrak{m}$ be its maximal ideal. The fibre product

$$R = A \times_E k = k + \mathfrak{m}$$

has normalization A, conductor $\mathfrak{m}$, and defect $A/R \simeq E/k$ as a k-vector space. After completed base change to E, it is the union of two formal n-planes meeting at their origins.

Choose $\alpha \in E \setminus k$. Then $A = R + R\alpha$, so A is finite, hence integral, over R. The two rings have the same fraction field because $\alpha = (\alpha x_1)/x_1$. Since A is normal, it is the normalization. An element of the conductor first lies in R, so its residue is some $c \in k$. Multiplication by the constants in E forces $cE \subset k$, hence $c = 0$; conversely $\mathfrak{m}A \subset \mathfrak{m}$. Thus the conductor is $\mathfrak{m}$, and reduction modulo $\mathfrak{m}$ gives $A/R \simeq E/k$ as k-vector spaces. Finally $E \otimes_k E \simeq E \times E$; the fibre-product description gives the two conjugate branches after completed base change.

The two projective three-spaces above meet transversely, so the completed local ring of Z is this pinching ring with $k = \mathbf{Q}$, $E = \mathbf{Q}(\sqrt{5})$, and $n = 3$. Its normalization, conductor, and defect $E/\mathbf{Q}$ are therefore forced by residue-field pinching; they do not depend on a six-variable presentation. Thus 5 is the discriminant square class of the two branches, not merely a value recovered from the sextic. This local model explains the descent geometry but does not replace the comparison with Hitchin's incidence scheme below. After base change to E, the pullback of the incidence double cover to $D(\sigma_3)$ is split. Galois conjugation exchanges both the two components and the two conjugate charts.

2.2 A rational incidence model

The function-field calculation requires a rational model, which can be given without descending either Clebsch chart. Let $W = \mathbf{Q}^3$ with the quadratic form Q from the introduction. For $a \in W$, infinitesimal rotation about a acts on H; put

$$\omega_a(p, q) = \langle a \cdot p, q \rangle.$$

These three rational skew forms define a closed subscheme

$$X = \{U \in \mathrm{Gr}(3, H) : \omega_a|_U = 0 \text{ for every } a \in W\}.$$

Hitchin identifies $X_{\mathbf{C}}$ with the Mukai–Umemura threefold [2, §§4–5]; in particular X is geometrically integral and smooth. The incidence scheme used here is

$$\mathcal{I} = \{([f], U) \in \mathbf{P}(H) \times X : \langle f, U \rangle = 0\}.$$

Let $\pi : \mathcal{I} \to \mathbf{P}(H)$ be the first projection. It is a projective bundle over X, is therefore normal and integral, and all its displayed equations are defined over $\mathbf{Q}$. Mukai and Umemura introduced the underlying threefold; Hitchin supplies this Grassmannian model and the incidence calculation [4, 2].

There is also a canonical link from the chart model to the finite incidence cover. Let

$$\mathcal{N} = \mathrm{Spec}_{\mathbf{P}(H)}\, \pi_* \mathcal{O}_{\mathcal{I}}.$$

Stein factorization gives $\mathcal{I} \to \mathcal{N} \to \mathbf{P}(H)$. Since $\mathcal{I}$ is normal and the degree calculation below makes its generic field quadratic over $\mathbf{Q}(\mathbf{P}(H))$, $\mathcal{N}$ is the normalization of $\mathbf{P}(H)$ in that field. For the icosahedron I_t, let $U_t \in X(E)$ be Hitchin's isotropic three-plane. The polarization identity and Hitchin's description give [1, §§2–5]

$$V_t = U_t^{\perp},$$

so $[p] \mapsto ([p], U_t)$ defines a section $\mathbf{P}(V_t) \to \mathcal{I}_E$. The two conjugate sections descend to a morphism

$$Z^\nu \longrightarrow \mathcal{I} \longrightarrow \mathcal{N}$$

over $\mathbf{P}(H)$. Over the finite etale neighborhood of $[xyz]$ constructed below, the local comparison proves that this morphism carries $\nu^{-1}([xyz]) = \mathrm{Spec}\, E$ isomorphically onto the golden incidence fibre.

2.3 Proof of Theorem 1.1

Hitchin realizes the Mukai–Umemura threefold as the zero locus of the three skew forms on $\mathrm{Gr}(3, H)$. The top Chern number of the dual universal bundle is 2, so a general f is orthogonal to two isotropic three-planes [2, §§4–5]. The incidence variety is a projective bundle over the integral Mukai–Umemura threefold, hence is integral; its generic function field is therefore a quadratic extension.

The reduced branch cycle.

Let $\mathcal{U}$ be the universal rank-three bundle on $\mathrm{Gr}(3, H)$, put $\mathcal{O}(1) = \det \mathcal{U}^*$, and write h for the hyperplane class on $\mathbf{P}(H)$ and for its pullback to $\mathcal{I}$. The threefold X is the smooth zero locus of the rank-nine bundle $\bigwedge^2 \mathcal{U}^* \otimes W$. Since

$$K_{\mathrm{Gr}(3,7)} = \mathcal{O}(-7), \qquad \det(\bigwedge^2 \mathcal{U}^* \otimes W) = \mathcal{O}(6),$$

adjunction gives $K_X = \mathcal{O}_X(-1)$. Put

$$\mathcal{F} = \ker(H \otimes \mathcal{O}_X \longrightarrow \mathcal{U}^*).$$

Then $\det \mathcal{F} = \mathcal{O}_X(-1)$, and, for the bundle projection $p : \mathcal{I} \to X$, one has $\mathcal{I} = \mathbf{P}_X(\mathcal{F})$, with its tautological $\mathcal{O}_{\mathcal{I}}(1)$ equal to $\pi^*\mathcal{O}_{\mathbf{P}(H)}(1)$. The projective-bundle canonical formula therefore gives

$$K_{\mathcal{I}} = \mathcal{O}_{\mathcal{I}}(-4) \otimes p^*(K_X \otimes \det \mathcal{F}^{-1}) = \pi^*\mathcal{O}_{\mathbf{P}(H)}(-4).$$

The determinant of $d\pi$ is consequently a nonzero section of $K_{\mathcal{I}} \otimes \pi^* K_{\mathbf{P}(H)}^{-1} = \pi^*\mathcal{O}(3)$. Its ramification divisor R has class $3h$. Because π has generic degree two, the projection formula gives the branch cycle

$$\pi_* R = 6h. \tag{2.1}$$

Here a tamely ramified prime in a quadratic extension has ramification index two and residue degree one, so each divisorial branch component occurs once in $\pi_* R$; components of R contracted to codimension at least two do not contribute.

Hitchin proves that a general real point of $J_0 = 0$ has exactly one incidence configuration, whereas either side has respectively two or zero real regular configurations [1, §9, Proposition 10 and Theorem 11]. At such a general boundary point π is quasi-finite, hence finite after shrinking the target. The source is smooth and the target regular of the same dimension, so miracle flatness makes this local finite map flat of degree two. Its fibre has only one geometric point and therefore is ramified. The real smooth points used here are Zariski dense in the irreducible sextic $J_0 = 0$. Thus this sextic is a component of the branch cycle. Equation (2.1) shows that it exhausts that cycle, with multiplicity one. In particular the normal quadratic incidence field is branched exactly along the reduced divisor $J_0 = 0$. This scheme-theoretic conclusion is the paper-local ramification calculation; the cited real classification supplies the dense set of boundary points, not the finite-cover assertion.

Square class from one fibre.

The fibre over $[xyz]$ is the model case announced in the introduction: its residue algebra will determine the constant twist of the global quadratic field. The first step below is an instance of a standard descent fact for quadratic covers, which we isolate because it is the reason that a single arithmetic fibre pins down a global twist. Let X be a normal, geometrically integral variety over a field k of characteristic not two, with $H^0(X, \mathcal{O}_X) = k$ and divisor class group free of two-torsion. Let $\mathcal{L}$ be a line bundle, let J be a nonzero section of $\mathcal{L}^{\otimes 2}$ whose zero divisor is irreducible, and let $M/k(X)$ be a quadratic extension branched exactly along that divisor. Then

$$M = k(X)(\sqrt{cJ})$$

for a square class $c \in k^\times/k^{\times 2}$ that is uniquely determined, in the shorthand of the introduction. Moreover, if $x \in X(k)$ lies off the branch divisor, then the normalization of X in M has residue algebra

$$k(\sqrt{c\,[J(x)]})$$

at x, where $[J(x)] \in k^\times/k^{\times 2}$ is the square class of the value $J(x)$ in the fibre $\mathcal{L}_x^{\otimes 2}$; this class is well defined because changing a basis of $\mathcal{L}_x$ rescales the value by a square. In particular, at a point where J takes a square value, a residue algebra $k(\sqrt{d})$ gives $c = d$.

Writing $M = k(X)(\sqrt{e})$, the odd valuations of e occur exactly along the branch divisor, so $\operatorname{div}(e/j)$ is even for $j = J/s^2$ with s a rational section of $\mathcal{L}$; the resulting divisor class is two-torsion, hence trivial, so $e = cjg^2$ with c a global unit, that is a constant. Specialization at x then evaluates $c\,[J(x)]$. The point of the statement is its last clause: one unramified rational point at which the branch section takes a square value, and whose fibre is a quadratic field, reconstructs every rational quadratic twist with the given branch divisor. The comparison carried out below supplies exactly that square value, since $J_0(xyz) = (16/25)^2$.

Square class from the branch divisor. Write $K = \mathbf{Q}(\mathbf{P}(H))$. In characteristic zero every quadratic extension of K has the form $K(\sqrt{d})$. The prime divisors at which d has odd valuation are exactly the branch divisors. The preceding ramification-cycle calculation proves that their reduced union is the irreducible sextic $J_0 = 0$.

Choose a nonzero rational section s of $\mathcal{O}_{\mathbf{P}(H)}(3)$ and set

$$j_s = \frac{J_0}{s^2} \in K^\times.$$

Its odd valuations occur exactly along $J_0 = 0$, because the divisor of s^2 is even. Consequently

$$\operatorname{div}(d/j_s) = 2D$$

for a divisor D on $\mathbf{P}(H)$. The class of D is two-torsion. Since $\operatorname{Pic}(\mathbf{P}(H)) = \mathbf{Z}$, D is principal. Thus

$$d = c j_s g^2$$

for some $g \in K^\times$ and a unique $c \in \mathbf{Q}^\times/\mathbf{Q}^{\times 2}$. The incidence extension is therefore $K(\sqrt{cJ_0})$.

The local comparison at xyz. Hitchin's classification gives exactly the two distinct configurations I_t and I_{1-t} over $[xyz]$ [1, §§3, 7–9]. Thus π is quasi-finite over this point. Because π is proper, there is a Zariski neighborhood B of $[xyz]$ on which π is finite. Let $\widetilde{B}$ be the normalization of B in the quadratic generic field $K(\sqrt{cJ_0})$. The normal finite scheme $\pi^{-1}(B)$ is birational to $\widetilde{B}$, hence is isomorphic to it. The branch-cycle calculation above identifies the branch locus with the reduced divisor $J_0 = 0$. Since $J_0(xyz) \neq 0$, shrinking B away from this divisor makes the finite normalization etale. The two distinct geometric points therefore form the complete reduced scheme-theoretic fibre, whose residue algebra is

$$E = \mathbf{Q}[t]/(t^2 - t - 1) = \mathbf{Q}(\sqrt{5}).$$

Choose a local generator e of $\mathcal{O}_{\mathbf{P}(H)}(3)$ at $[xyz]$, and write $j_e = J_0/e^2$. The normalization is locally $w^2 = cj_e$. Changing e multiplies j_e by a unit square. Moreover, $j_e(xyz)$ differs from the displayed value $J_0(xyz) = (16/25)^2$ by a rational square. Specialization therefore gives the residue algebra $\mathbf{Q}(\sqrt{c})$. Comparison with the preceding fibre proves $c = 5$ in $\mathbf{Q}^\times/\mathbf{Q}^{\times 2}$. This also explains why evaluating the generic square class at the point is legitimate: the point lies in the finite etale locus and the change from j_s to j_e is a unit square there.

The Stein algebra. The quadratic field involution extends to the normalization $\mathcal{N}$. If $f : \mathcal{N} \to \mathbf{P}(H)$, then, since 2 is invertible,

$$f_*\mathcal{O}_{\mathcal{N}} = \mathcal{O} \oplus \mathcal{M}$$

is the decomposition into invariant and anti-invariant summands. The invariant summand is $\mathcal{O}$: its relative spectrum is finite and birational over the normal space $\mathbf{P}(H)$. Normality makes $f_*\mathcal{O}_{\mathcal{N}}$ an S_2 module over the regular base, and the direct summand $\mathcal{M}$ is therefore a rank-one reflexive sheaf. Every such sheaf on projective space is a line bundle, say $\mathcal{M} \simeq \mathcal{O}(-d)$.

Multiplication on the anti-invariant summand is a section of $\mathcal{M}^{-2} \simeq \mathcal{O}(2d)$. Its zero divisor is the branch divisor, which is the sextic $J_0 = 0$; hence $d = 3$, and multiplication is cJ_0 for some $c \in \mathbf{Q}^\times$. The function-field calculation and the golden fibre give $c = 5$ modulo a rational square. Rescaling the chosen isomorphism $\mathcal{M} \simeq \mathcal{O}(-3)$ absorbs that square and gives the exact algebra

$$f_*\mathcal{O}_{\mathcal{N}} \simeq \mathcal{O} \oplus \mathcal{O}(-3), \qquad z^2 = 5J_0.$$

After Hitchin's scale for J_0 and an isomorphism $\mathcal{M} \simeq \mathcal{O}(-3)$ giving multiplication $5J_0$ are fixed, the remaining choices differ by $z \mapsto -z$, the deck involution. Thus the equation used on the Clebsch chart below is an equation of finite algebras, not only of their function fields.

Integral boundary. Choose a full lattice $\Lambda \subset H$ and an integer $a \neq 0$ such that $J_{\mathbf{Z}} = a^2 J_0$ has integral coefficients. The algebra

$$\mathcal{O}_{\mathbf{P}(\Lambda)} \oplus \mathcal{O}_{\mathbf{P}(\Lambda)}(-3), \qquad z^2 = 5J_{\mathbf{Z}},$$

is finite locally free of degree two over $\mathbf{Z}$ and has the function field of Theorem 1.1. On $D(10J_{\mathbf{Z}})$ it is finite etale. The primes 2 and 5 are forced bad primes for an ordinary two-sheet interpretation: in characteristic 2 the equation is inseparable, while in characteristic 5 it is generically $z^2 = 0$.

This abstract integral algebra does not by itself give an integral incidence theorem. The constructions in the cited Mukai–Umemura and Hitchin sources are over $\mathbf{C}$, and the available binary-sextic formulae justify their classical invariant presentation only after inverting 30. Spreading out the rational incidence isomorphism gives some $\mathbf{Z}[1/N]$, with $2, 3, 5 \mid N$ after enlarging N, but neither the sources nor the present equations determine the other prime divisors or prove that there are none. Consequently every finite-field icosahedral-incidence interpretation in this paper remains restricted to an unspecified cofinite set of primes. The formula for the abstract quadratic algebra itself has no such extra exception.

The missing assertion is precise. One must identify an integral harmonic lattice and its three-skew-form Grassmannian zero locus with an integral Mukai–Umemura model, prove flatness and normality, and show that Stein formation and the chart sections commute with the required base changes. Until those geometric steps are supplied, extra divisors of N are evidence gaps rather than additional bad primes predicted by the quadratic equation.

3 The marked orientation source

This section keeps track of the sign lost by projectivizing the Clebsch chart. The normalization is intrinsic to the pullback. Its comparison with the golden six-axis data and the Petersen coefficient module is relative to the marked bridge datum $\mathfrak{m}$ fixed in the introduction.

3.1 Normalization over the Clebsch chart

Let $\mathcal{N} \to \mathbf{P}(H)$ be the incidence normalization from Section 2, and put $B = \mathbf{P}(V_E)$. Choose the anti-invariant generator in Theorem 1.1; its pullback is the odd generator z below. On the chart, the Stein-algebra theorem and Hitchin's restriction identity give the equality of finite $\mathcal{O}_B$-algebras

$$z^2 = 5\iota_t^* J_0 = 80\sigma_3^2 = (z - 4\sqrt{5}\,\sigma_3)(z + 4\sqrt{5}\,\sigma_3).$$

For a normal domain S, with 2 invertible and $0 \neq a \in S$, the algebra $S[z]/(z^2 - a^2)$ maps to $S \times S$ by $z \mapsto (a, -a)$. Its image consists of pairs congruent modulo a; hence its normalization is $S \times S$, its conductor is $(a) \times (a)$, and it is already split etale where a is a unit. Applying this elementary pinching lemma with $a = 4\sqrt{5}\,\sigma_3$ gives the normalization

$B_+ \amalg B_-$ and locates the entire failure of normality on the Clebsch cubic. The constant plane U_t supplies one component; deck exchange supplies the other. Hitchin's chart classification identifies $D(\sigma_3)$ as exactly the locus of two distinct regular configurations. The magnitude $4\sqrt{5}$ is forced by the product of the descent class 5 and the chart factor 16; choosing its sign is precisely choosing one normalized component.

The sign of z is the remaining freedom in the chosen anti-invariant generator. We normalize it by declaring the component supplied by U_t to have $z = 4\sqrt{5}\,\sigma_3$. This does not strengthen the integral statement: comparison with the geometric incidence model still holds only over an unspecified $\mathbf{Z}[1/N]$.

3.2 The golden involution

Choose the displayed normalized representatives $a_0, \dots, a_5$ of the ordered projective axes in $\mathfrak{m}$, and write $C = C_{\mathfrak{m}}$ and $Z_C = Z_{\mathfrak{m}}$. Their Gram matrix has the form

$$G = (t+2)I + tC, \qquad C = \begin{pmatrix} 0 & 1 & 1 & 1 & -1 & -1 \\ 1 & 0 & -1 & -1 & -1 & -1 \\ 1 & -1 & 0 & 1 & 1 & -1 \\ 1 & -1 & 1 & 0 & -1 & 1 \\ -1 & -1 & 1 & -1 & 0 & -1 \\ -1 & -1 & -1 & 1 & -1 & 0 \end{pmatrix}.$$

The six vertex axes of an icosahedron as equiangular lines, and their signed Gram or Seidel-matrix description up to diagonal switching, are classical [12, §3]. What is tracked below is the sign of the specific marked triangle cubic under the golden exchanger. The frame operator $\sum_i a_i a_i^{\mathsf{T}}$ commutes with the irreducible three-dimensional A_5-action, so it is scalar. Its trace is $6(t+2)$; division by the dimension gives $\sum_i a_i a_i^{\mathsf{T}} = 2(t+2)I_3$. Hence $G^2 = 2(t+2)G$. Substituting $G = (t+2)I + tC$ and using $(t+2)^2 = 5t^2$ gives $C^2 = 5I$. Changing signs of axis representatives switches C to DCD, but every triangle product and hence Z_C is unchanged. The off-diagonal entries of C^2 also give

$$\sum_{k \neq i,j} C_{ij} C_{jk} C_{ki} = 0.$$

Thus the directional derivative of Z_C along $\mathbf{1}$ vanishes, so $Z_C(x + u\mathbf{1}) = Z_C(x)$ and the cubic descends to $\mathbf{Q}^6/\mathbf{Q1}$.

Let a_i' denote the conjugate axes, obtained by $t \mapsto 1 - t$, and let R be the exchanger of Section 4. Direct substitution gives

$$Ra_i = t d_i a'_{p(i)}, \qquad p = (0, 1, 5, 4, 2, 3), \qquad (d_i) = (1, -1, 1, 1, 1, 1).$$

Using $t^2(1-t) = -t$, one obtains

$$d_i d_j C_{p(i)p(j)} = -C_{ij}.$$

Thus, at the golden fibre, conjugation transported by R changes C to $-C$ and Z_C to $-Z_C$. The lone negative d_i is only a change of axis representatives and does not cause the reversal; switching alone leaves the triangle cubic fixed.

No global marking of the varying configurations on B_- is needed or asserted. The fixed section U_t selects B_+, but does not by itself choose $\mathfrak{m}$. After $\mathfrak{m}$ is fixed, we attach its constant source $[C_{\mathfrak{m}}, Z_{\mathfrak{m}}]_{\mathrm{or}}$ to B_+ and attach the deck-opposite source to B_-, meaning that the normalized odd generator changes sign. The calculation at $[xyz]$ proves that this relative convention agrees with the actual conjugate golden configuration there; it is not used to infer a pointwise conference labeling elsewhere. Since the normalized components remain disjoint above $\sigma_3 = 0$, the two relative source labels extend there even though the literal two-configuration interpretation fails.

3.3 Transport to degree six

Write Hitchin's chart as $\iota_t(y) = \sum_i y_i q_i^{(t)}$, with $q_1 = xyz$. Since $R(xyz) = -xyz$, equivariance and irreducibility of the four-module give

$$Rq_i^{(t)} = -q_i^{(1-t)}$$

after transporting the five plane-triple labels. Normalize the lift on B_+ by $q_1 = xyz$; the deck-opposite source carries the lift $-\iota_t$. The displayed exchanger calculation shows that this normalization agrees with golden conjugation at $[xyz]$. Thus the lift is part of the oriented source datum, rather than a purported global marking of the second family.

The primitive integral map

$$\beta(y)_{ij} = y_i + y_j$$

lands in the Petersen (-2)-eigenspace and satisfies

$$\sum_{i<j}(y_i + y_j)^2 = 3\sum_i y_i^2, \qquad y_i = \frac{1}{3}\sum_{j \neq i} \beta(y)_{ij}.$$

Multiplicity one makes this comparison unique after the five labels and the normalization are fixed. Reversing the chart lift reverses y, the zonal harmonic F_y, and its cubic integral. The two signs in the normalized incidence cover therefore select the two signs of the exact Gaunt cubic relative to $\mathfrak{m}$. This proves Proposition 1.2.

The remaining marking actions and the boundary of this comparison are listed in Appendix A.

4 Arithmetic specialization

The arithmetic exchange is already visible on the cubic xyz. Its two icosahedral six-axis configurations are Galois conjugate over $\mathbf{Q}(\sqrt{5})$. The exchanger is the projective orthogonal transformation carrying one configuration to the other. Over a finite field where the configurations become rational, it has spinor class $[2]$; modulo 11 this class is nontrivial.

4.1 The golden fibre

Retain the field E, parameter t, and six-set I_t from Section 2. These six projective points are its axes through opposite vertices. Thus an icosahedral configuration here means the unordered six-set I_t. A point of the incidence cover records one chosen six-set; its two points over xyz correspond to the two conjugate configurations.

Proposition 4.1 (Golden fibre)**.** *The points of I_t lie on $xyz = 0$, have a common nonzero norm, have normalized squared inner product $1/5$, and form a six-arc. The two configurations are conjugate geometric configurations defined over*

$$\mathbf{Q}[t]/(t^2 - t - 1) = \mathbf{Q}(\sqrt{5})$$

and are exchanged by its nontrivial automorphism. In Hitchin's incidence cover, whose point records the chosen six-set, they are the complete fibre over $[xyz]$.

Proof. Every displayed vector has one zero coordinate. Since $t^2 = t + 1$, its squared norm is $t^2 + 1 = t + 2$. The inner product of two distinct representatives is $\pm t$: the A_5-action is transitive on pairs of distinct projective axes, so one squared inner-product check suffices. Moreover

$$(t + 2)^2 = 5(t + 1) = 5t^2.$$

This gives the normalized squared inner product $1/5$. The twenty three-point checks reduce to one signed Gram determinant. For any triple, let $\epsilon = \pm 1$ be the product of its three inner-product signs. Its Gram determinant is

$$(t + 2)^3 - 3(t + 2)t^2 + 2\epsilon t^3 = \begin{cases} 4t^4, & \epsilon = 1, \\ 4t^2, & \epsilon = -1. \end{cases}$$

It is nonzero, so every triple is independent. The final assertion uses Hitchin's classification of the two icosahedral sets on the orthogonal-plane cubic. □

The six-arc conclusion remains valid after reduction in every characteristic other than 2. The metric interpretation additionally excludes characteristic 5, where $t^2 - t - 1$ is inseparable and the common norm $t + 2$ vanishes. This does not extend the complete-fibre assertion: the comparison with Hitchin's geometric incidence cover is available in characteristic zero and, after spreading out, only away from an unspecified finite set of primes.

4.2 The exchanger

Set

$$R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

The relation $1 - t = -1/t$ gives, by substitution,

$$R(I_t) = I_{1-t}, \qquad R^2 = \mathrm{diag}(1, -1, -1), \qquad R^4 = 1.$$

Moreover $R(xyz) = -xyz$: it fixes the projective cubic and exchanges its two configurations.

4.3 Finite-field specialization

Let k be a finite field of odd characteristic other than 5, and suppose that 5 is a square in k. Then $t^2 - t - 1$ has two distinct roots in k, so the displayed equations give two rational configurations exchanged by R.

Proposition 4.2 (Spinor specialization)**.** *The spinor class of R is $[2] \in k^\times/k^{\times 2}$. Under the standard identifications*

$$\mathrm{SO}_3(k)/\Omega_3(k) \cong \mathrm{PGL}_2(k)/\mathrm{PSL}_2(k),$$

the exchanger lies outside $\mathrm{PSL}_2(k)$ exactly when 2 is nonsquare. In particular, over $\mathbf{F}_{11}$ it represents the nontrivial element of

$$T_{11} = \mathrm{PGL}_2(11)/\mathrm{PSL}_2(11).$$

Proof. For $Q(x, y, z) = x^2 + y^2 + z^2$, reflection in a vector v has spinor class $Q(v)$. On the yz-plane,

$$R = s_{e_2} s_{e_2 - e_3},$$

so

$$\theta(R) = Q(e_2)Q(e_2 - e_3) = 2 \quad \text{in } k^\times/k^{\times 2}.$$

The standard identifications

$$\mathrm{SO}_3(k) \cong \mathrm{PGL}_2(k), \qquad \Omega_3(k) \cong \mathrm{PSL}_2(k)$$

identify the spinor quotient with the projective quotient. Finally, the roots modulo 11 are 4 and 8, and 2 is nonsquare modulo 11, proving the specialization. □

The class [2] is forced by the orthogonal conjugacy class of this quarter-turn on the yz-plane; the two reflections merely evaluate its spinor norm.

Thus one equivariant quadratic datum carries two independent characters. The degree-two parameter algebra $k[t]/(t^2 - t - 1)$, with deck involution $t \mapsto 1 - t$, has discriminant class [5]; its two configurations are rational exactly when that torsor splits. The rational orthogonal element R realizes the deck involution on the configurations and has spinor character [2]. Dye's existence criterion for Clebsch hexagons has the same [5] boundary [3, Theorem 1].

Integral boundary. Over a finite field of characteristic different from 2 and 5, the fibre $\mathcal{Q}_f$ of the abstract quadratic cover $z^2 = 5J(f)$ over a rational point f gives

$$\#\mathcal{Q}_f(\mathbf{F}_q) = 1 + \chi_q(5J(f)).$$

Its geometric interpretation over finite fields is proved only away from an unspecified finite set of primes. This global comparison is separate from the good reduction at 11 of the explicit golden fibre and matrix R; we do not replace the finite exceptional set by $p \neq 2, 5$.

5 The golden operator and its cubic shadows

The incidence cover supplies a golden orientation bit. Relative to the marked bridge datum, we attach that bit to the six-axis carrier and apply three elementary operations to its conference operator: middle exterior power, commutator with the diagonal algebra, and centered squaring. These operations produce the Joubert, Segre, Igusa, and diagonal Clebsch models in one calculation. The main route first fixes the outer family and identifies four exact descriptions of its marked cubic. A reader following the source–shadow–return argument may then continue directly to Section 6. The later exchange-spectrum and reconstruction subsections prove independent structural consequences of the same conference carrier.

Let X be the ordered six-axis set and $A_X = \mathbf{Q}^X/\mathbf{Q}\mathbf{1}$. The outer six-set $\mathcal{T}$ consists of the six synthematic totals, that is, the six labelled one-factorizations of the fixed labelled graph K_X. These are the classical mystic pentagons. For each $T \in \mathcal{T}$, its complementary triangle colouring assigns signs $\epsilon_T(S) \in \{\pm 1\}$ to the three-subsets $S \subset X$, with the global sign fixed by the marked outer orientation. The signs obey the four-point two-graph identity and transform by

$$\epsilon_{gT}(gS) = \mathrm{sgn}(g)\epsilon_T(S) \qquad (g \in S_X).$$

Writing $\epsilon_T(S) = (-1)^{\tau_T(S)}$, the four-point identity is

$$\sum_{S \in \binom{Q}{3}} \tau_T(S) = 0 \qquad \left(Q \in \binom{X}{4}\right). \tag{5.2}$$

This is the two-graph parity equation. Switching a graph representative leaves τ_T unchanged; complementing τ_T reverses every triangle sign while preserving (5.2). This is the signed coloured-triangle construction of the outer six-family [13, Sections 1.1–1.2 and 1.6]. The order-six conference two-graph is likewise classical: its switching class has the pentagon-plus-isolated-vertex form and automorphism group A_5 [14]. The four-point identity reconstructs from each ϵ_T a switching class of symmetric zero-diagonal sign matrices; it is the unique order-six conference class. Choose representatives C_T whose triangle products are ϵ_T, and transport the Hodge orientations with the ordered axis basis. Then $C_T^2 = 5I$. These choices constitute the coherent outer marking. The incidence sheet supplies only its orientation bit, not the outer labels or representatives.

Put

$$J_T(x) = \sum_{|S|=3} \epsilon_T(S) x_S, \qquad x_S = \prod_{i \in S} x_i.$$

The following table fixes every sign. Take T_0 to be the total attached to the matrix displayed in Section 3, let $T_r = p_r T_0$, and write permutations in one-line notation. In the column headed "coefficient word", the signs occur in the order

$$012, 013, 014, 015, 023, 024, 025, 034, 035, 045, 123, 124, 125, 134, 135, 145, 234, 235, 245, 345.$$

r	p_r	coefficient word of J_{T_r}
0	012345	−−+++−++−−++−−+−−−++
1	012354	++−−−−++−+−+−−++++−−
2	012435	+−+−+−−−++−−+++−+−+−
3	012453	−+−+++−−−+−+++−−−+−+
4	012534	−++−−++−+−+−+−−++−−+
5	012543	+−−+−+−++−+−−+−+−++−

(5.1)

Indeed, $\epsilon_{pT_0}(S) = \operatorname{sgn}(p)\epsilon_{T_0}(p^{-1}S)$; applying this to the twenty triangle products of the displayed matrix gives the six rows. They are the signed coloured-triangle Joubert frame of [13, Sections 1.1–1.2 and 1.6]. Thus this citation supplies no unstated scalar or sign convention: its six coordinates are exactly the forms J_T in (5.1).

5.1 Four cubic shadows

For a three-subset $S \subset X$, set

$$K_T = *\bigwedge^3 C_T, \qquad Z_T(x) = \frac{1}{4} \sum_{|S|=3} (K_T)_{SS} x_S, \qquad D_x = \operatorname{diag}(x).$$

The Hodge star is taken in the ordered axis basis. Writing $X = \{0, \ldots, 5\}$ in its order and S^c for the complementary triple in increasing order,

$$*e_S = \operatorname{sgn}(S, S^c)\, e_{S^c}, \qquad \operatorname{sgn}(S, S^c) = (-1)^{\sigma(S)-3}, \qquad \sigma(S) = \sum_{i \in S} i,$$

where $\operatorname{sgn}(S, S^c)$ is the sign of the permutation listing S and then S^c. The exponent counts that listing's inversions: writing $S = \{a_0 < a_1 < a_2\}$, the element a_k has k elements of S below it, hence $a_k - k$ of S^c, and these sum to $\sigma(S) - 3$. Since $\sigma(S) + \sigma(S^c) = 15$, complementary signs multiply to $(-1)^9 = -1$, which is $*^2 = -1$ in middle degree. The orientation therefore travels with the ordered basis: an odd relabelling or switching transports both K_T and the sign of its diagonal, and it is not legitimate to switch the matrix while holding the old Hodge convention fixed. Figure 2 separates the four normalized formulas from their classical consequences.

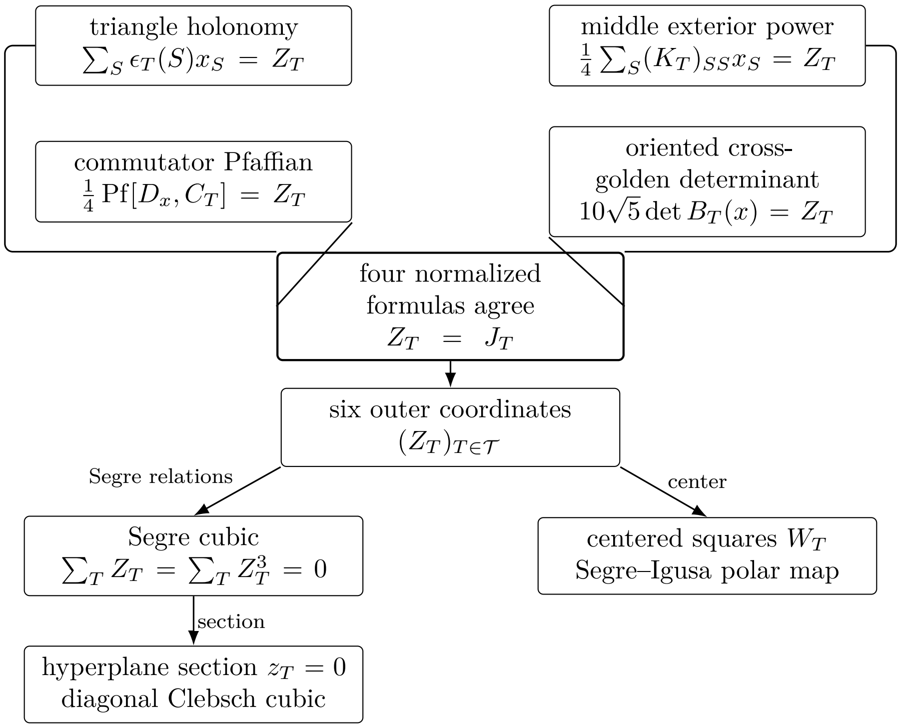

Figure 2: The four constructions of the same marked cubic and their classical targets. Plain connectors denote the normalized identities displayed in the upper boxes; arrows denote subsequent constructions. The determinant-line orientation is normalized by the cross-golden formula.

What the four descriptions do and do not assert.

The four formulas below are not four coincidences of equal standing, and it is worth separating them before proving them. For *every* symmetric matrix A, expanding $\mathrm{Pf}[D_x, A]$ over perfect matchings makes the coefficient of x_S the signed complementary minor $\det A[S^c, S]$, which the Hodge convention fixed above identifies with $(*\bigwedge^3 A)_{SS}$. So the middle-exterior diagonal and the commutator Pfaffian are one identity written twice, and their agreement records the orientation convention rather than any property of C. The cross-golden determinant presupposes the splitting into golden eigenspaces, so once $C^2 = 5I$ is known it is a reformulation; its content is that $B_T(x)$ is defined without choosing bases, so its determinant is a map of determinant lines. The remaining agreement is not formal. The complementary minor $\det A[S^c, S]$ is a polynomial in the entries joining S to its complement, whereas the triangle product $a_{ij}a_{jk}a_{ki}$ uses only entries inside S; for a general symmetric zero-diagonal matrix the two are unrelated. That they are proportional here is a property of the conference class, and it is the one identification among the four that could fail.

For a six-tuple $(a_U)_{U\in\mathcal{T}}$, write

$$\mathrm{center}_T\, a_T = a_T - \frac{1}{6}\sum_{U\in\mathcal{T}} a_U.$$

Theorem 5.1 (Golden operator and classical cubic shadows)**.** *For a marked bridge datum and its coherent outer family:*

1. *The triangle products of C_T satisfy*

$$(K_T)_{SS} = 4(C_T)_{ij}(C_T)_{jk}(C_T)_{ki} = 4\epsilon_T(S) \quad (S = \{i,j,k\}),$$

so $Z_T = J_T$ is the switching-invariant orientation cubic on A_X. The four-point two-graph identity and pair balance recover the conference switching class; after fixing a vertex, the negative edges on the other five vertices form a pentagon.

2. *The same cubic is a commutator Pfaffian:*

$$\mathrm{Pf}[D_x, C_T] = 4Z_T(x), \qquad \det[D_x, C_T] = 16Z_T(x)^2.$$

Over $E = \mathbf{Q}(\sqrt{5})$, put $P_{T,\pm} = (I \pm C_T/\sqrt{5})/2$ and $B_T(x) = P_{T,-}D_xP_{T,+}$. At T_0, orient the two determinant lines so that the following equality has positive sign, and transport those orientations through the coherent outer marking. Then

$$Z_T(x) = 10\sqrt{5}\det B_T(x).$$

3. *The six coordinates $(Z_T)_{T\in\mathcal{T}}$ satisfy*

$$\sum_T Z_T = 0, \qquad \sum_T Z_T^3 = 0.$$

They are the signed Joubert coordinates and hence define the classical map from six labelled points to the Segre cubic

$$\left\{[z] \in \mathbf{P}^5 : \sum_T z_T = \sum_T z_T^3 = 0\right\}.$$

Inside its hyperplane $z_T = 0$, the remaining linear relation cuts the ambient space to $\mathbf{P}^3$, and the remaining cubic equation is the diagonal Clebsch cubic surface.

4. *The centered determinants*

$$W_T = Z_T^2 - \frac{1}{6}\sum_U Z_U^2 = \frac{1}{16}\operatorname{center}_T \det[D_x, C_T]$$

satisfy

$$\sum_T W_T = 0, \qquad \left(\sum_T W_T^2\right)^2 = 4\sum_T W_T^4.$$

Thus W is the Segre–Igusa polar map.

Proof. The tight-frame argument in Section 3 gives $C_T^2 = 5I$. Its off-diagonal entries are the pair-balance identities

$$\sum_{k\neq i,j} (C_T)_{ij}(C_T)_{jk}(C_T)_{ki} = 0.$$

Triangle products are unchanged by diagonal switching and obey the four-point identity because every edge of a four-set occurs twice. In the gauge $(C_T)_{0i} = 1$, pair balance says that every vertex among the other five has two negative incident edges. The negative graph is therefore a five-cycle. This directly recovers the classical switching-class normal form without an enumeration.

Expand the Pfaffian of $[D_x, C_T]$ over perfect matchings. The coefficient of x_S pairs S bijectively with S^c, and its signed sum is the complementary 3-minor of C_T. The Hodge convention displayed above makes that signed minor equal to $(K_T)_{SS}$. There are two triangle orbits, distinguished by $\epsilon_T(S) = \pm 1$; switching, transported relabelling, and complementation reduce them to $S = 012$ and $S = 014$ in the displayed matrix. For these representatives the complementary blocks $C_T[S^c, S]$ are

$$\begin{pmatrix} 1 & -1 & 1 \\ -1 & -1 & 1 \\ -1 & -1 & -1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & -1 & 1 \\ 1 & -1 & -1 \\ -1 & -1 & -1 \end{pmatrix}.$$

Their determinants are 4 and -4; the Hodge signs $\operatorname{sgn}(S^c, S)$ are both -1, while the internal triangle products are -1 and $+1$. Hence in both orbits $(K_T)_{SS} = 4(C_T)_{ij}(C_T)_{jk}(C_T)_{ki}$. This exposes the complementary-minor step rather than treating it as formal. The Pfaffian is therefore $4Z_T$; squaring gives the determinant identity.

The projectors split the axis space into two three-spaces. Relative to that splitting,

$$[D_x, C_T] = \begin{pmatrix} 0 & -2\sqrt{5}\, B_T(x)^\mathsf{T} \\ 2\sqrt{5}\, B_T(x) & 0 \end{pmatrix}.$$

Comparison with $16Z_T^2$ gives $Z_T^2 = 500 \det(B_T)^2$; the chosen determinant-line orientation fixes the displayed sign.

The matching expansion gives $Z_T = J_T$ coefficient by coefficient. Table (5.1) therefore identifies the six operator cubics with the signed Joubert coordinates, including their normalization and the rule $gZ_T = \operatorname{sgn}(g) Z_{gT}$. Their Segre relations are classical [13, Sections 2.1–2.2], so $\sum Z_T = \sum Z_T^3 = 0$. Setting one coordinate to zero gives $\sum_{U \neq T} Z_U = \sum_{U \neq T} Z_U^3 = 0$, the diagonal Clebsch cubic. The centered-square formula and the symmetric Igusa equation are the classical Segre–Igusa duality map [13, Section 2.2], following van der Geer. Here the determinant identity in the theorem identifies that polar vector with the centered commutator determinants. □

Why the determinant. The map $B_T(x) : V_{T,+} \to V_{T,-}$ is defined without choosing bases. Its determinant is therefore a map between determinant lines and becomes the displayed scalar once those lines are oriented. A permanent has no analogous basis-free meaning. Indeed, changing orthonormal frames sends a matrix B to $R_-^\mathsf{T} B R_+$; already for $B = I_3$, right multiplication by a rotation through angle θ in a coordinate plane changes its permanent from 1 to $\cos(2\theta)$. Thus the permanent supplies no further cubic shadow of the marked operator.

No physical or exceptional-group interpretation is used here. The retained chain is incidence descent, conference carrier, classical cubics, and the harmonic return of Section 6.

For the main source–shadow–return route, continue with Section 6. The next two subsections use the same conference carrier but supply no hypothesis for Theorem 6.1. The switching recovery inside Theorem 5.1 concerns the fixed order-six conference class; the later reconstruction theorem is a global faithfulness statement for aligned four-set data.

5.2 Exchange spectra of symmetric conference matrices

The next theorem is a general spectral formula for every symmetric conference order; order six is its answer, not its hypothesis. It explains why this six-axis carrier has no cut-independent higher-order analogue. Let $C = C^\mathsf{T}$ be any symmetric conference matrix of order $2d$, that is, a zero-diagonal matrix with off-diagonal entries in $\{\pm 1\}$ and $C^2 = qI$, where $q = 2d - 1$. Its zero trace gives two eigenspaces of dimension d. For a balanced half Y, order the coordinates by (Y, Y^c) and write

$$C = \begin{pmatrix} A & R \\ R^\mathsf{T} & E \end{pmatrix}.$$

Let D_Y be the diagonal sign involution of this cut, put $P_\pm = (I \pm C/\sqrt{q})/2$, and choose an isometry $Q_+ : \mathbf{R}^d \to \operatorname{im} P_+$. The balanced exchange operator is

$$H_Y = Q_+^\mathsf{T} D_Y P_- D_Y Q_+.$$

Theorem 5.2 (Balanced exchange rigidity)**.** *With the notation above,*

$$\operatorname{Spec}(H_Y) = \operatorname{Spec}\left(\frac{1}{q} RR^\mathsf{T}\right) = \left\{1 - \frac{\alpha_1^2}{q}, \dots, 1 - \frac{\alpha_d^2}{q}\right\},$$

where $\alpha_1, \dots, \alpha_d$ are the eigenvalues of A. The spectrum is independent of the balanced half Y if and only if $d \leq 3$. Thus order six is the unique nontrivial realized symmetric conference order with this property, and its exchange spectrum is

$$\left\{\frac{1}{5}, \frac{4}{5}, \frac{4}{5}\right\}.$$

For a four-set K, let $w(K)$ be the sum of its three signed Hamilton-cycle products. Call K aligned *if $w(K) = 3$; in standard two-graph terminology this is the union of the coherent and incoherent four-set types. Let c_Y count the aligned four-sets in Y. Then*

$$\operatorname{tr}(H_Y) = \frac{d^2}{q}, \qquad \operatorname{tr}(H_Y^2) = \frac{F_d + 32c_Y}{q^2},$$

where

$$F_d = dq^2 - 2qd(d-1) + d(d-1) + 12\binom{d}{3} - 8\binom{d}{4}.$$

Hence the first exchange moment is universal, whereas the second is cut-independent exactly in the stated small orders.

Proof. Put $Q = C/\sqrt{q}$ and $L = [D_Y, Q]$. The commutator anticommutes with Q, and on its positive eigenspace one has $Lv = 2P_- D_Y v$. In cut coordinates,

$$L = \frac{2}{\sqrt{q}} \begin{pmatrix} 0 & R \\ -R^\mathsf{T} & 0 \end{pmatrix}.$$

Consequently $4H_Y$ is L^*L restricted to the positive eigenspace. Thus H_Y has the squared singular values of $R/\sqrt{q}$. The Y-principal block of $C^2 = qI$ is

$$RR^\mathsf{T} = qI - A^2,$$

which proves the spectral formula. Complementary principal-block spectral relations of this kind are classical [15].

Since A is a zero-diagonal sign matrix, $\operatorname{tr}(A^2) = d(d-1)$, giving the first trace formula. Sorting the closed four-walks by support gives

$$\operatorname{tr}(A^4) = d(d-1) + 12\binom{d}{3} + 8 \sum_{\substack{K \subset Y \\ |K|=4}} w(K).$$

Every edge occurs twice in the product of the three cycle signs, so $w(K) \in \{3, -1\}$. Substitution yields the displayed formula for $\operatorname{tr}(H_Y^2)$. This is the exact finite version of the closed-walk organization used for conference-frame subensembles in [16].

Suppose now that $d \geq 4$ and even the second exchange moment is independent of Y. Then the sums of the aligned-four-set indicator over all balanced halves are equal. The characteristic-zero inclusion matrix from four-sets to d-sets has full column rank [17, Theorem 1]; hence the indicator itself is constant on four-sets. Switch C so that all edges incident with one vertex ∞ are positive. On $\{\infty, i, j, k\}$, if the remaining edge signs are x, y, z, then

$$w = xy + xz + yz.$$

If the common value is 3, every triangle on the other $2d - 1$ vertices is monochromatic. All its edges have one sign, making the inner product of two corresponding conference rows equal to $2d - 2$, a contradiction. If the common value is -1, that two-colouring has no monochromatic triangle; the elementary equality $R(3, 3) = 6$ gives $2d - 1 \leq 5$, again a contradiction.

For $d = 1$ the conclusion is immediate, and for $d = 2$ every possible two-by-two principal sign block squares to I. For $d = 3$, if τ is the product of the three edge signs, then

$$A^2 = 2I + \tau A,$$

and $\det(tI - A) = t^3 - 3t - 2\tau$. This polynomial factors as $(t - 2)(t + 1)^2$ or $(t + 2)(t - 1)^2$, so A^2 always has spectrum $\{4, 1, 1\}$. The spectral formula then gives $\{1/5, 4/5, 4/5\}$. $\square$

The obstruction is local, but the aligned sets retain uniform aggregate statistics. Greaves and Suda proved that the determinant-(-3) four-subsets of a symmetric conference matrix form a 3-$(2d, 4, (d - 3)/2)$ design [18, Example 2.3]; the identity $\det C[K] = 3 - 2w(K)$ identifies their blocks with the aligned sets above. The regular-two-graph parameters give the same identification [19, Section 2.2 and Proposition 4.1]. Classical Johnson-scheme inclusion calculus [20, Section 8] therefore gives, for a uniformly chosen balanced half and $d \geq 4$,

$$\mathbf{E}c_Y = \rho\binom{d}{4}, \qquad \mathrm{Var}(c_Y) = \frac{\binom{2d}{4}\binom{2d-8}{d-4}}{\binom{2d}{d}}\rho(1 - \rho), \qquad \rho = \frac{d - 3}{2(2d - 3)}.$$

These formulas also hold modulo complementation because $c_Y = c_{Y^c}$. At order ten they give $\mathbf{E}c_Y = 5/7$ and $\mathrm{Var}(c_Y) = 10/49$, hence $\mathbf{E}[c_Y(c_Y - 1)] = 0$. Since c_Y is a nonnegative integer, it is therefore 0 or 1 on every cut. The 126 projective cuts consequently split into 36 with $c_Y = 0$ and 90 with $c_Y = 1$. The known design fixes this second-moment split without making the complete higher-order spectrum uniform.

5.3 Reconstructing the signing

The design still retains the full discrete signing, despite fixing only aggregate low moments. Let $\tau : \binom{V}{3} \to \mathbf{F}_2$ be a two-graph on a finite set V, meaning that it satisfies (5.2) with X replaced by V. Let $\mathcal{A}(\tau)$ be the four-sets on which the four values of τ are all equal. Complementing τ preserves this family.

Equation (5.2) is the definition of a two-graph, and the correspondence between two-graphs and switching classes of graphs, used below as the two-graph equation, is likewise standard: two-graphs are due to Higman, with Taylor the standard reference and Seidel's switching the graph-side description; see Brouwer and Van Maldeghem [26, §1.1.12] for both, with attributions.

Theorem 5.3 (Aligned-design faithfulness)**.** *If $|V| \geq 7$, then $\mathcal{A}(\tau)$ determines τ up to complement, and seven is the least bound with this property.*

For a symmetric conference matrix of order $n \geq 10$, its marked family of principal four-subsets with determinant -3, which is the design of Greaves and Suda [18, Table 1 and

Example 2.3], therefore determines its signing up to diagonal switching and global negation. After one aligned four-set Q is known, the

$$1 + 4(n-4) + 6\binom{n-4}{2} = 3n^2 - 23n + 45$$

tests on four-sets meeting Q in at least two points suffice, and the decoder in the proof uses exactly this many. An aligned anchor is found deterministically with at most twenty tests. Each of those queries reads one bit, whether a single fourth-order principal minor equals -3; the reconstruction consumes no minor value and no minor of any other order. One calibrated triangle product selects between the two global orientations.

Proof. Fix $r \in V$ and form a graph G_r on $V \setminus \{r\}$, with ij an edge when $\tau(rij) = 1$. The two-graph equation gives

$$\tau(ijk) = \tau(rij) + \tau(rik) + \tau(rjk).$$

Thus $\{r, i, j, k\}$ is aligned exactly when ijk is a clique or an independent triple of G_r. The equality $R(3,3) = 6$ supplies an aligned anchor on every seven-set.

It remains to prove faithfulness on seven points. Fix an aligned $Q = \{1, 2, 3, 4\}$, complement one candidate if necessary, and switch graph representatives so that both vanish on Q. Switch each outside vertex once more so that its edge to vertex 4 is zero. An outside point x then determines a uniquely normalized cut

$$p_x = (e_{1x}, e_{2x}, e_{3x}, e_{4x}) \in \mathbf{F}_2^4, \qquad (p_x)_4 = 0.$$

The four tests meeting Q in three points have the following complete signature table; an entry ijk means that $\{i, j, k, x\}$ is aligned.

p_x	aligned triples	p_x	aligned triples
0000	$123, 124, 134, 234$	1000	234
0100	134	0010	124
1110	123	$1100, 1010, 0110$	$\varnothing$

Thus they recover p_x, except that the three balanced cuts $B_{12} = 1100$, $B_{13} = 1010$, and $B_{14} = 0110$ have the same empty signature.

We record the remaining finite step explicitly. For outside points x, y, put $p = p_x$, $s = p_y$, $e = e_{xy}$, and let $\mathcal{B}(p, s, e)$ be the set of pairs $ij \subset Q$ for which $\{i, j, x, y\}$ is aligned. Its four triangle values give

$$ij \in \mathcal{B}(p, s, e) \quad \Longleftrightarrow \quad p_i + s_i = p_j + s_j \ \text{ and } \ e = p_j + s_i. \tag{5.3}$$

This criterion gives a complete classification. If p and s are already known, some pair has $p_i + s_i = p_j + s_j$, and exactly one value of e makes that pair aligned. If exactly one cut is balanced, relabelling Q reduces the known cut to 0000 or 1000; substituting the three balanced cuts in (5.3) gives

p	s	$\mathcal{B}(p,s,0)$	$\mathcal{B}(p,s,1)$
0000	B_{12}	34	12
0000	B_{13}	24	13
0000	B_{14}	14	23
1000	B_{12}	34	$13, 14$
1000	B_{13}	24	$12, 14$
1000	B_{14}	$12, 13$	23

so both the balanced cut and e are recovered. When both cuts are balanced, the remaining table is

$\{p, s\}$	$\mathcal{B}(p, s, 0)$	$\mathcal{B}(p, s, 1)$
$\{B_{12}, B_{12}\}$	$12, 34$	$13, 14, 23, 24$
$\{B_{13}, B_{13}\}$	$13, 24$	$12, 14, 23, 34$
$\{B_{14}, B_{14}\}$	$14, 23$	$12, 13, 24, 34$
$\{B_{12}, B_{13}\}$	23	14
$\{B_{12}, B_{14}\}$	13	24
$\{B_{13}, B_{14}\}$	12	34

The rows are distinct. The sole loss of information is the order of two distinct balanced cuts: interchanging p and s preserves the six pair tests and leaves e fixed.

There are three outside points. Suppose one cut changes between the two candidates. For each of the other two points the classification forces the pair to consist of distinct balanced cuts and swaps them. Those two other cuts are therefore equal to the same changed cut. Their mutual pair cannot be a distinct-cut swap, so it must agree pointwise, contradicting either of the first two swaps. No cut changes, and then (5.3) recovers every outside edge. The candidates agree after the initial possible complement.

For larger V, apply the seven-point result to every seven-set. Adjacent seven-sets in the connected Johnson graph share six vertices. On any triple in that intersection the two restrictions cannot choose opposite complement conventions, so their complement choices agree. This proves faithfulness. This proof is also the promised decoder. Fix a root and six other vertices and test the twenty four-sets formed by the root and a triple of the six; $R(3, 3) = 6$ supplies an aligned anchor Q. Query all four-sets meeting Q in at least two points. First reconstruct one seven-set $Q \cup \{x, y, z\}$. For each remaining vertex u, apply the same seven-point tables to $Q \cup \{x, y, u\}$; agreement on the fixed six-set $Q \cup \{x, y\}$ selects the same complement convention and recovers the cut of u. Once all cuts are known, (5.3) recovers every outside edge. No further query is used, so the number of selected tests is exactly the quadratic count in the theorem.

For a conference signing, the identity $\det C[Q] = 3 - 2w(Q)$ identifies the determinant-(-3) family with $\mathcal{A}(\tau)$. Choosing a root and switching its incident signs to one recovers every matrix entry from the rooted triangle values. The sole remaining complement sends C to $-C$, and one triangle product fixes that orientation. Minimality of the bound seven is the six-point pair of Remark 5.4. □

Remark 5.4 (Seven points is sharp). Six points do not suffice, and not only because a two-graph may have no aligned four-set at all. Represent a two-graph on $\{0, \ldots, 5\}$ by the unique graph in its switching class in which 0 is isolated, and take

$$G = \{12,\ 15,\ 24,\ 25,\ 35\}, \qquad H = \{13,\ 14,\ 24,\ 34,\ 35\}.$$

Both two-graphs have aligned family exactly $\{\{0, 1, 2, 5\}, \{0, 1, 3, 4\}\}$, and H is neither G nor its complement. So $\mathcal{A}$ does not determine a two-graph on six points up to complement, and the hypothesis $|V| \geq 7$ of Theorem 5.3 cannot be weakened.

Remark 5.5 (What the query count is measured against). For a general symmetric matrix, Holtz and Sturmfels identify the principal-minor ambiguity as diagonal sign similarity under their strict nondegeneracy hypothesis [27, Thm. 6]. No such hypothesis is needed here: for a Seidel matrix the order-three minors are twice its triangle signs, so they recover the switching class directly. Against an oracle returning minor *values* the problem is solved in $O(n^2)$ queries and that count is asymptotically optimal [28, 29]; for a Seidel matrix the cycle-basis algorithm needs only the $\binom{n}{2} - n + 1$ principal minors of order three [30], each of which is twice a triangle sign and so returns the two-graph outright. The count above improves on none of that and is not offered as an algorithm. It is what a decoder pays when it is given only the fourth-order indicator, which is invariant under global negation as the order-three data is not.

Two lower bounds hold for that data. Two-graphs on n points form an $\mathbf{F}_2$-space of dimension $\binom{n}{2}-n+1$; each test returns one bit and determination up to complement allows fibres of size two, so any family of tests — and any decision tree, by its leaf count — has at least $\binom{n}{2}-n$ members. Families fixed in advance obey a stronger bound, because the answers are biased. A uniformly random two-graph restricts to a uniformly random two-graph on any four points, and two of those eight are aligned, so every alignment test answers yes with probability exactly $1/4$. The information-theoretic bound of combinatorial search theory, subadditivity of entropy applied to the entropy $\binom{n-1}{2}-1$ of the complement pair, which is what the answers determine, gives

$$k \;\geq\; \frac{\binom{n-1}{2}-1}{H(1/4)} \;>\; 1.2326\Big(\tbinom{n-1}{2}-1\Big) \;\approx\; 0.616\,n^2,$$

where $H(p)=-p\log_2 p-(1-p)\log_2(1-p)$, for every family of k tests that determines the two-graph up to complement. The ratio of the exhibited count $3n^2-23n+45$ to this floor tends to 4.87, and the constant 3 is not known to be optimal.

The floor does not apply to a decoder that chooses each test in the light of earlier answers, and for large n such a decoder does better than any fixed family can, using $\binom{n}{2}+O(n)$ tests — the counting bound to leading order. Read the two-graph on seven points, which costs at most the $\binom{7}{4}$ tests available there, and then add one point at a time. Attaching a point v to six points carrying a known restriction costs a bounded number of tests, because the twenty tests meeting v inside those seven points determine its five rooted values: the fifteen tests avoiding v are already known, and the theorem separates on seven points. Every further rooted value at v then costs a single test. Indeed, for a new old vertex u, the two hypothetical values of $\tau(ruv)$, together with the already recovered data, define two candidate restrictions on any seven-set containing u, v and five of the six anchor vertices. They agree on the anchor five-set and hence are not complements. By the seven-point theorem some alignment test distinguishes them. Every test not containing both u and v is already decided, so one of the tests through u, v and two anchors distinguishes the two values. The decoder evaluates both hypothetical outcomes, selects such a test, and its one-bit answer fixes $\tau(ruv)$. So the price of the coherence restriction is a price of fixing the tests in advance: adaptively the fourth-order indicator matches the order-three minor values to leading order, both at $n^2/2$.

Tests indexed by four-sets occur elsewhere as a primitive observation. The causal-clustering algorithm of Kummerfeld and Ramsey [31] decides one coherence bit per quartet of observed variables, up to $\binom{n}{4}$ of them, each by a pair of statistical tests on sampled data. That bit reports whether covariance minors vanish under a one-factor measurement model, not whether four triple signs cohere, so the two counts measure different work and are not compared here.

The same statement in principal minors.

For a Seidel matrix the theorem is a statement about principal minors, and it is worth recording which ones it uses. The principal minors of order at most three trivialize the problem: the 1×1 minors vanish, the 2×2 minors are all -1, and the third-order minors return the two-graph outright, as Remark 5.5 records. The theorem consults none of them. Its data are the fourth-order principal minors, through $\det C[Q]=3-2w(Q)$ below; that these take only the two values -3 and 5 is Greaves and Suda's Table 1, and the -3 fibre is their design. They are invariant under global negation, so they cannot distinguish C from $-C$ and see each four-set only through the single bit $\det C[Q] \in \{-3, 5\}$. The content is that this negation-invariant fourth-order data still determines the matrix up to exactly that ambiguity, that seven vertices are enough and six are not by Remark 5.4, and that a quadratic family of these determinants suffices.

This theorem belongs to the reconstruction-up-to-complementation literature. On four vertices a two-graph has zero, two, or four triples, so equality of aligned families is precisely four-hypomorphy up to complementation within the class of two-graphs. Dammak, Lopez, Pouzet, and Si Kaddour prove the analogous four-local reconstruction theorem for ordinary graphs [24]. For arbitrary 3-uniform hypergraphs, Pouzet and Si Kaddour prove that the least local size yielding eventual reconstruction is five [25]. The two-graph parity axiom therefore lowers that general threshold to four, with ambient order seven sharp. That sharpness is a separate statement from Dammak, Lopez, Pouzet, and Si Kaddour's: their $v \geq 7$ is the endpoint of the admissible range $4 \leq k \leq v - 3$, and the sharpness they remark on is in k, so no six-point failure follows from their work for either observable. The homogeneous-triple analysis of ordinary graphs [23] supplies the one-anchor viewpoint; the away-from-anchor tests above remove its balanced-cut ambiguity.

We have not located this sharp two-graph specialization in the bounded audit. Row `OPER-4` of the public claim–proof–novelty ledger records the search depth and access gaps; the paper claims no priority for the general hypomorphy framework or the homogeneous-triple method.

The neighboring framework is spectral monomorphy of principal submatrices [21, 22]. Here only the squared spectrum is assumed constant. We have not located the balanced singular-spectral classification above in that literature; the block identity, closed-walk expansion, inclusion rank, and aligned design are credited as indicated. Row `OPER-3` of the same public ledger records the bounded search scope.

This exchange invariant deliberately forgets signs: it sees A^2, or equivalently RR^{T}. The cubic stage above retains the determinant-line orientation. Cut-independent squared singular spectrum, determinant orientation, and the later cubic shadows are therefore distinct claims.

6 The degree-six harmonic realization

Here $\mathcal{H}_6$ denotes the real degree-six spherical harmonics on S^2. This section is the harmonic return of the diagonal Clebsch invariant isolated by the operator theorem, not another operator shadow. The marked pair-sum map carries the relative orientation to the Petersen coefficient module, after which the construction passes from icosahedral face axes to $\mathcal{H}_6$.

Put $\phi = (1+\sqrt{5})/2$. The following vectors have squared norm 3; their projective classes are the ten axes through opposite faces of an icosahedron:

$$\begin{aligned} &v_{12} = (-1,-1,-1), &&v_{13} = (-1,-1,1), &&v_{14} = (-1,1,-1), &&v_{15} = (-1,1,1),\\ &v_{34} = (0,-\phi^{-1},-\phi), &&v_{25} = (0,-\phi^{-1},\phi), &&v_{45} = (-\phi^{-1},-\phi,0),\\ &v_{23} = (-\phi^{-1},\phi,0), &&v_{35} = (-\phi,0,-\phi^{-1}), &&v_{24} = (\phi,0,-\phi^{-1}). \end{aligned} \tag{1}$$

We use the unit representatives $u_{ij} = v_{ij}/\sqrt{3}$. Two labels are adjacent when the pairs are disjoint. This is the Petersen graph $KG(5,2)$, and the labeling is geometric because, for distinct axes,

$$(u_{ij} \cdot u_{kl})^2 = \begin{cases} 5/9, & \{i,j\} \cap \{k,l\} = \varnothing,\\ 1/9, & \text{otherwise.} \end{cases}$$

Replacing ϕ by its conjugate gives the conjugate embedded configuration. The abstract Petersen incidence and the formulas below are unchanged after the corresponding relabeling. We do not claim that this displayed embedding of its four-space is defined over $\mathbf{Q}$.

The zonal degree-six harmonics define

$$L : \mathbb{R}^{10} \longrightarrow \mathcal{H}_6, \qquad L(a)(\omega) = \sum_{i<j} a_{ij} P_6(u_{ij} \cdot \omega),$$

where

$$P_6(s) = \frac{1}{16}\left(231s^6 - 315s^4 + 105s^2 - 5\right).$$

We equip $\mathcal{H}_6$ with the normalized spherical inner product

$$\langle f, g\rangle = \frac{1}{4\pi}\int_{S^2} f(\omega)g(\omega)\, d\omega.$$

Theorem 6.1 (Degree-six harmonic restriction)**.** *Let u_{ij}, $1 \le i < j \le 5$, be the explicitly labelled unit axes through opposite icosahedral faces in* (1)*, and set*

$$F_y(\omega) = \sum_{i<j}(y_i + y_j)P_6(u_{ij} \cdot \omega).$$

The coefficient vectors $(y_i + y_j)_{i<j}$ form the Petersen (-2)-eigenspace, and their zonal harmonics give an injective copy of V in the real degree-six spherical-harmonic space $\mathcal{H}_6$. The spherical cubic on this copy is

$$\frac{1}{4\pi}\int_{S^2} F_y(\omega)^3\, d\omega = -\frac{784000}{1247103}\sigma_3(y).$$

6.1 Proof of Theorem 6.1

Let A be the Petersen adjacency matrix in the labeling (1). Direct evaluation gives the reproducing-kernel matrix

$$K_{ef} = P_6(u_e \cdot u_f) = \frac{1}{243}(196I + 47J - 112A).$$

Indeed $P_6(1) = 1$, while the two off-diagonal orbit values are $P_6(\sqrt{5}/3) = -65/243$ and $P_6(1/3) = 47/243$. Thus the three matrix coefficients contain no further data: $196 + 47 = 243$ on the diagonal and $47 - 112 = -65$ on Petersen edges. The addition theorem for spherical harmonics gives

$$G_{ef} = \langle P_6(u_e \cdot -), P_6(u_f \cdot -)\rangle = \frac{1}{13}K_{ef}.$$

Thus G, not K, is the Gram matrix for the stated inner product. Only two off-diagonal evaluations occur: A_5 is transitive on pairs of two-subsets with intersection zero and with intersection one. The three eigenvalues below therefore follow from the Petersen eigenvalues $3, -2, 1$, rather than from a ten-by-ten diagonalization. Its eigenvalues on $\mathbb{R}^{10} \simeq \mathbf{1} \oplus V_4 \oplus V_5$ are respectively

$$\frac{110}{1053}, \qquad \frac{140}{1053}, \qquad \frac{28}{1053}.$$

They are positive, so L is injective.

The Petersen eigenvalues are $3, -2, 1$ on $\mathbf{1}, V_4, V_5$. For $a_{ij} = y_i + y_j$ with $\sum_i y_i = 0$,

$$(Aa)_{ij} = \sum_{\{k,l\}\cap\{i,j\}=\varnothing}(y_k + y_l) = -2(y_i + y_j).$$

These vectors form the four-dimensional (-2)-eigenspace.

This identifies the abstract bridge uniquely. The action of A_5 on five letters is two-transitive, so an endomorphism of $\mathbf{Q}^5$ commuting with A_5 has one constant diagonal entry and one constant off-diagonal entry. Its restriction to V is therefore scalar. Consequently, once the five plane triples are identified, every A_5-equivariant comparison between the arithmetic coordinate module and the Petersen coefficient module is a scalar multiple of

$$y \longmapsto (y_i + y_j)_{i<j}.$$

The inverse on its image is

$$y_i = \frac{1}{3}\sum_{j \neq i} a_{ij}.$$

Requiring the comparison to preserve σ_3 forces the scalar to have cube one, hence to equal one over $\mathbf{Q}(\sqrt{5})$. The abstract comparison is therefore unique with the stated labels and normalization. It still supplies no map between the two ambient harmonic spaces. The linear lift selected in Proposition 1.2 fixes its sign.

Nor can a rotation-equivariant polynomial covariant induce this linear bridge. Suppose such a map $\Phi : H_{\mathbb{R}} \to \mathcal{H}_6$ restricts to the displayed Clebsch four-space as the nonzero linear comparison above. Decomposing Φ into homogeneous parts and applying it to λy shows that its degree-one part would have the same restriction. But $\mathrm{Hom}_{SO_3}(H_{\mathbb{R}}, \mathcal{H}_6) = 0$, because these are nonisomorphic irreducible rotation modules. Thus the marked A_5-intertwiner is exactly the level at which a linear comparison can exist; no ambient rotation-equivariant polynomial covariant restricts to it.

The cubic integral is A_5-invariant on V_4. The invariant cubic line is generated by σ_3, so one nonzero value determines the scalar. The vector $y = (4, -1, -1, -1, -1)$ spans the fixed line of a point stabilizer. Relative to the marked labels, this primitive vector fixes the scale and sign of the one-vector normalization. Its quadratic norm needs no new moment calculation: pair-sum tightness and the V_4 Gram eigenvalue give

$$\frac{\langle F_y, F_y \rangle}{\sum_i y_i^2} = 3\left(\frac{140}{1053}\right) = \frac{140}{351}, \qquad \frac{1}{4\pi}\int F_y^2 = \frac{2800}{351}.$$

Only the cubic normalization remains. Exact evaluation on this fixed line gives

$$\frac{1}{4\pi}\int F_y^3 = -\frac{15680000}{1247103}, \qquad \sigma_3(y) = 20,$$

which proves the coefficient in Theorem 6.1. This one normalization value follows by expanding P_6 and applying

$$\frac{1}{4\pi}\int_{S^2} \omega_1^{2a}\omega_2^{2b}\omega_3^{2c}\, d\omega = \frac{(2a-1)!!(2b-1)!!(2c-1)!!}{(2a+2b+2c+1)!!};$$

monomials with an odd exponent integrate to zero.

The Gaunt factorization.

The apparently large coefficient has two independent sources:

$$-\frac{784000}{1247103} = -\left(\frac{400}{46189}\right)\left(\frac{1960}{27}\right).$$

The first factor is $\left(\begin{smallmatrix} 6 & 6 & 6 \\ 0 & 0 & 0 \end{smallmatrix}\right)^2$. Thus $46189 = 11 \cdot 13 \cdot 17 \cdot 19$ is the universal degree-six Wigner denominator. The second factor is the one-vector restriction scalar after the unit-axis and primitive pair-sum normalizations have been fixed. Multiplicity one is what reduces the whole cubic calculation to this marked fixed line; $1960/27$ is therefore a normalization scalar, not an experimental coincidence. These denominator primes arise from the stated spherical and pair-sum normalizations and are unrelated to the incidence-cover square class. In particular the prime 11 occurring here is the degree-six angular-momentum denominator, and it has no connection with the field of order eleven over which the Clebsch code is reconstructed elsewhere in this series; the coincidence of the two elevens is accidental. In the quadratic identity above, $3 = \|\beta(y)\|^2/\|y\|^2$, the second factor is the spherical Gram eigenvalue on V_4, and the factor 13 in $1053 = 13 \cdot 81$ is $\dim \mathcal{H}_6$.

Remark 6.2. In the orthonormal Condon–Shortley convention, write $F = \sum_{m=-6}^{6} q_{6m} Y_{6m}$ and

$$W_6(F) = \sum_{m_1+m_2+m_3=0} \begin{pmatrix} 6 & 6 & 6 \\ m_1 & m_2 & m_3 \end{pmatrix} q_{6m_1} q_{6m_2} q_{6m_3}.$$

The Gaunt formula and $\left(\begin{smallmatrix} 6 & 6 & 6 \\ 0 & 0 & 0 \end{smallmatrix}\right) = -20/\sqrt{46189}$ give

$$\int_{S^2} F^3 \, d\omega = -\frac{130}{\sqrt{3553\pi}} W_6(F).$$

This is the standard unnormalized degree-six bond-order invariant [6, 7, 8]. Combining this conversion with Theorem 6.1 gives the exact restriction in that standard language:

$$W_6(F_y) = \frac{313600\pi^{3/2}}{4563\sqrt{3553}} \sigma_3(y).$$

Here $3553 = 46189/13$ and $4563 = 27 \cdot 13^2$; both are forced by converting the universal Gaunt factor from the normalized spherical inner product to the standard Condon–Shortley convention.

7 Conclusion

Hitchin's two conjugate Clebsch charts live over $\mathbf{Q}(\sqrt{5})$, where the branch sextic becomes $16\sigma_3^2$. Their common rational point $[xyz]$ lies in the finite etale locus of the rational incidence model. Its complete reduced fibre has residue algebra $\mathbf{Q}(\sqrt{5})$ and determines the remaining square class, giving the exact quadratic algebra

$$\mathcal{O} \oplus \mathcal{O}(-3), \qquad z^2 = 5J_0.$$

After the marked bridge datum and determinant-line orientations are fixed, the chosen sheet selects the oriented conference source or its opposite. The sheet alone supplies neither the axis ordering and outer labels nor the chart lift and Petersen labels. Triangle holonomy, the diagonal of $*\bigwedge^3 C$, the commutator Pfaffian, and the cross-golden determinant are then the same marked cubic. Its six outer translates give the signed Joubert coordinates, the Segre cubic and its diagonal Clebsch sections, and the Segre–Igusa polar map. The Petersen pair-sum construction carries the relative orientation to the degree-six zonal model, where the normalized spherical cubic restricts to $-784000\sigma_3/1247103$. This compares orientation sources relative to the marking: $Z_{\mathfrak{m}}$ and σ_3 live on different spaces, and no map between their ambient harmonic representations is asserted.

The same carrier has two independent consequences. Order six is the unique nontrivial symmetric conference order whose balanced exchange spectrum is cut-independent. From seven vertices onward, aligned four-sets reconstruct a two-graph up to complement; for conference matrices this recovers every signing from order ten onward up to switching and global negation. These results describe what the carrier remembers without supplying a hypothesis to the source–shadow–return comparison.

The explicit configurations specialize at 11 without a global integral incidence model there. Determining the exact finite set of primes that must be inverted for the geometric incidence comparison remains the integral identification, flatness, normality, and Stein-base-change problem isolated in Section 2; it does not affect the characteristic-zero quadratic algebra above. Thus one fully marked golden source has four exact operator descriptions, its outer coordinates recover the classical cubic models, and its Petersen realization returns an exact multiple of the Clebsch invariant σ_3.

A Marking ambiguities

The relative formulation uses the following complete action of the auxiliary choices.

operation	effect
axis switching	$C \mapsto DCD$, while every triangle product and Z is fixed; this is the equivalence already built into the marked source.
axis relabelling	simultaneously permutes the rows and columns of C and the variables of Z; the proposition transports equivariantly and does not choose an order.
five-label relabelling	simultaneously permutes the q_i, the coordinates y_i, and the two-subset labels; σ_3 is fixed and β is equivariant. Relabelling only one side changes $\mathfrak{m}$.
chart scaling	rescales the linear identification and the transported cubic generator while leaving the projective chart fixed. It is not quotiented: $q_1 = xyz$ fixes the normalization. In particular, scaling by -1 reverses the generator.
golden Galois conjugation	after transport by R, sends C, Z, and the normalized chart lift to their negatives at the golden fibre.
incidence deck exchange	interchanges B_+ and B_- and negates the odd generator; relative to $\mathfrak{m}$, the attached source and harmonic generator are negated by definition.

Thus switching and coordinated relabelling do not change the comparison. Lift normalization and the cross-identification of the two five-labelled systems remain visible inputs. Proposition 1.2 makes no descent claim for them.

There is no nonzero rotation-equivariant linear comparison between harmonic degrees three and six: they are nonisomorphic irreducible rotation modules of dimensions seven and thirteen. The linear comparison used here is the unique marked A_5-intertwiner on the chosen Clebsch four-space, and it stops at the relative marked source class $[C_{\mathfrak{m}}, Z_{\mathfrak{m}}]_{\mathrm{or}}$.

B Reproducibility and data availability

The square-class argument and both specialization propositions are proved in the text from Hitchin's incidence calculation and boundary classification, the paper-local ramification-cycle calculation, and the displayed equations. A paper-local exact-arithmetic bundle independently checks the canonical-degree bookkeeping, golden configurations, exchanger, and spinor decomposition. The human six-arc proof uses one Gram-determinant formula indexed by the triangle sign, so the stored twenty minors are regression data rather than twenty proof obligations.

A second paper-local bundle reconstructs the ten axes over $\mathbf{Q}(\sqrt{5})$, distinguishes the reproducing kernel from the normalized spherical Gram matrix, verifies the Petersen labeling, and evaluates the quadratic and cubic moments. Its independent replay uses a separate dense polynomial representation and checks the Wigner $3j$ conversion. Neither calculation uses floating-point arithmetic. The proof uses symmetry to reduce the kernel matrix to its two Petersen relations and uses only one point-stabilizer-fixed vector to normalize the invariant cubic line.

A third bundle checks, for the displayed marking, the conference square, the golden exchanger and its transport of C and Z_C, the scalar factorization of the pulled-back cover, all six signed outer coefficient words, and the primitive Petersen pair-sum identities. It reuses the canonical harmonic certificate only through its pinned digest and exact witness values. Normalization of the pulled-back incidence scheme and its extension across the branch divisor remain human arguments.

The operator theorem has a separate structural route. Its conference, triangle, two-graph, middle-exterior square and diagonal, support recovery, and golden-descent mechanisms are checked by the pinned golden-return Lean gate. Middle-degree Hodge complementation is not checked entry by entry there: complementation on the triple basis is proved to be set complement on the underlying three-element subsets, the sign attached to a triple is proved to be the sign of its concatenation with the increasing complementary triple, and the square identity then follows from the parity of two complementary label sums. The same gate proves symbolically that the fixed conference commutator Pfaffian is four times its triangle cubic and checks the three-vertex signed-matrix identity used in the order-six exchange-spectrum converse. The unrestricted inclusion-rank and Ramsey exclusion is a human proof. Theorem 5.3 is kernel-checked in its two-graph form: for every two-graph on a finite set of at least seven points, the aligned four-sets determine the triangle values on distinct triples up to one global complement bit. The formal proof follows the one above, except in the last step: instead of descending along adjacent seven-sets of the Johnson graph it places any two triples in a single seven-point restriction and calibrates the complement bit against a fixed triple. Both bounds behind the triangle Ramsey equality $R(3,3) = 6$ and the aligned four-set they produce through a root are checked; so are the passage from arbitrary labels to the normalized cut coordinates, the seven-point cut signatures with their distinct-balanced-cut ambiguity, the symbolic elimination of that ambiguity by the third outside point, the distinctness of the seven points, the extension of any two triples to a common seven-point restriction, and the agreement of the complement bits on the overlap. Lean also checks the conference-signing switching transport, the identity $\det C[Q] = 3 - 2w(Q)$, and the count of twenty possible anchor tests. The selected query family is defined there as the four-point subsets meeting a fixed four-point anchor in at least two points, and its cardinality is checked to be $3n^2 - 23n + 45$, and those tests alone are checked to determine the two-graph up to the same global complement bit. For a Seidel matrix Lean also checks that a principal four-by-four minor on distinct labels is -3 exactly on the aligned four-sets and 5 otherwise, which identifies the marked determinant-(-3) family with the aligned family, and that two such matrices on at least seven labels with the same determinant-(-3) family differ by a diagonal switching and one global sign. The deterministic anchor search enters formally as the existence of an aligned four-set among the twenty tested and the count of those tests, not as a procedure. Coherence of the outer six-family, the cross-golden determinant comparison, and the classical Joubert–Segre–Igusa identifications remain human proofs with primary literature inputs; they are not inferred from a finite matrix check. The sharpness of the seven-point hypothesis is proved in the text by the six-point pair of Remark 5.4, and the query bounds of Remark 5.5 are proved there from the theorem and the answer marginal alone. Neither is a Lean statement, and neither is used in the proof of Theorem 5.3.

The `verification` directory distributed with the paper contains the exact programs, canonical certificates, checksums, toolchain requirements, and aggregate replay command. It also records the exact claim-to-evidence correspondence. Three paper-specific Lean gates now prove the abstract pinching and conductor mechanisms, involutive splitting, golden discriminant and reflection characters, tight-frame conference mechanism, marked relative sign transport, Petersen eigenspace and two-orbit kernel calculation, and fixed-line cubic normalization.

No step of any of the three gates leaves the kernel. Their finite content is discharged by kernel decision over an explicitly bounded domain, by rewriting, or by cofactor expansion; every audited terminal depends only on `propext`, `Classical.choice` and `Quot.sound`, and several on strictly fewer. Compiled evaluation occurs nowhere in the pinned source closures, and each replay refuses it, so this is a checked property of the distributed artifact rather than a declared boundary. Its declaration map records partial mechanism coverage only: no complete geometric manuscript row is declared formally proved at its full geometric strength. In particular, the global incidence and Stein identifications, scheme-theoretic chart correspondence, geometric golden-fibre identification, face-axis addition theorem, and raw spherical moment remain human inputs. The exact-arithmetic audits remain independent checks rather than hidden proof steps.

AI assistance disclosure

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